

Ghadeer Al Jufout

(562) 509-9183 | galjufout@gmail.com | [linkedin.com/in/ghadeer03](https://www.linkedin.com/in/ghadeer03) | github.com/Ghadeer122003

EDUCATION

University of California, Irvine
B.S. in Software Engineering

June 2026
GPA: 3.98

SKILLS

Languages: Python, JavaScript, C++, MATLAB

Backend: Node.js, Express, FastAPI, REST APIs, JWT, PostgreSQL, Prisma

Frontend: React (Vite, Router), HTML, CSS

Cloud & Infrastructure: AWS (EC2, S3, Lambda, Elastic IP), Docker, Kubernetes, Nginx

CI/CD & DevOps: GitHub Actions, Microservices, Distributed Systems, Container Orchestration

RESEARCH EXPERIENCE

IEEE GESS 2024 – UC Irvine

2024

Undergraduate Researcher, Distributed Systems & Machine Learning

Long Beach, CA

Paper [\[Link\]](#) | Code [\[Link\]](#)

- Published “MATLAB-Based Lightweight Workload Prediction via ML in Distributed Systems” at IEEE GESS 2024; indexed on IEEE Xplore (DOI: 10.1109/GESS2024).
- Designed and built an AutoML model selector (Linear Regression, SVR, Random Forest) that auto-routes workloads to the optimal algorithm, achieving a **37% reduction in runtime latency** across 3 dataset tiers.
- Deployed across 3 Docker-containerized distributed nodes (**54% faster** than centralized) and integrated a 4-worker MATLAB parallel pool, cutting CPU time by **62%** on large datasets.

EXPERIENCE

UCI Campus Reserve [\[Link\]](#)

2025–2026

Software Engineering Intern, Full-Stack

Irvine, CA | *Python, FastAPI, React, AWS EC2, Nginx*

- Built a full-stack automated ML pipeline on a team of engineers to classify UCI Campus Reserve wildlife images at scale using SpeciesNet; system capable of processing **100k+ image backlog** via AWS GPU instances.
- Developed a stats dashboard and image upload interface; connected frontend to FastAPI backend, enabling end-to-end data flow from upload to classification results, processing **2,341 images in under 7 minutes** at 5.75 img/sec (~**10× faster** than local, which took 11–20 min for 500 images).
- Engineered a resume/retry mechanism for failed batch jobs, ensuring fault-tolerant processing with zero data loss on pipeline errors; reduced manual intervention for failed runs.

PROJECTS

WildFire Tracker [\[Link\]](#)

2026

DevOps & Full-Stack Engineer

React, Node.js, Express, PostgreSQL, Prisma, AWS EC2, JWT

- Led all cloud deployment:** provisioned AWS EC2 with Elastic IP and Nginx reverse proxy, built the full production pipeline for a real-time wildfire tracking app aggregating CAL FIRE, NASA FIRMS, and NWS data.
- Engineered CI/CD-ready environment setup and production build pipeline; ensured zero-downtime deployments across the full stack.
- Built JWT-protected REST API endpoints with Node.js/Express including session management; integrated email alerts via Resend API and PWA-ready offline support pages with React/Vite.

P2P File Sharing System [\[Link\]](#)

2025

Systems Engineer

Python, Docker, Socket Programming, Distributed Systems

- Architected from scratch** a decentralized P2P file transfer system over LAN supporting **10+ concurrent peers** using custom socket programming and a peer discovery protocol with real-time conflict resolution.
- Designed and configured multi-peer topology via Docker Compose across 5+ isolated container nodes, cutting setup time by **60%**; implemented zero-data-loss node failure handling via persistent manifests and checksum verification.
- Built a custom binary transfer protocol with chunked file splitting, enabling reliable large-file transfers and partial download recovery across unstable LAN connections.